

Set Theory for Computer Science

Week 1: Sets

Sets: definition and notation

Fundamental set operations

Venn diagrams and partitions

The algebra of sets

Definition and notation of sets

A **set** is an unordered collection of elements

Examples

DaysOfWeek := { *Mon, Tue, Wed, Thu, Fri, Sat, Sun* }

A := { 1, 2, 3 }

Digits := { 0, 1, ..., 9 }

$\mathbb{N}$:= { 0, 1, 2, 3, ... }

natural numbers

$\mathbb{Z}$:= { ..., -2, -1, 0, 1, 2, ... }

integer numbers

Definition and notation of sets

A **set** is an unordered collection of elements

Examples

$DaysOfWeek := \{Mon, Tue, Wed, Thu, Fri, Sat, Sun\}$

$A := \{1, 2, 3\}$

$Digits := \{0, 1, \dots, 9\}$

$\mathbb{N} := \{0, 1, 2, 3, \dots\}$

natural numbers

$\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}$

integer numbers

Prototype notation

$MultiplesOf2 := \{2k : k \text{ a natural number}\} = \{0, 2, 4, \dots\}$

$Months := \{x : x \text{ is a month}\}$

Element and subset

Element

$a \in A$	means	a is an element of the set A
$a \notin A$	means	a is not an element of the set A

Subset

$A \subseteq B$	means	all elements of A are elements of B
$A \not\subseteq B$	means	at least one element of A is not in B

Element and subset

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Subset

$A \subseteq B$	means	all elements of A are elements of B
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Examples

$4 \in \{1, 2, 3, 4\}$	$\{2, 3\} \subseteq \{1, 2, 3, 4\}$
$5 \notin \{1, 2, 3, 4\}$	$\{2, 5\} \not\subseteq \{1, 2, 3, 4\}$

Set equality and the empty set

Definition of set equality

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

Examples

$$\{1, 2, 3, 4\} = \{4, 3, 1, 2\} = \{4, 3, 3, 2, 1, 2\}$$

$$\{1, 2, 3, 4\} \neq \{2, 3, 4, 5\}$$

Set equality and the empty set

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Examples

$$\{1, 2, 3, 4\} = \{4, 3, 1, 2\} = \{4, 3, 3, 2, 1, 2\}$$

$$\{1, 2, 3, 4\} \neq \{2, 3, 4, 5\}$$

Definition of empty set

$\emptyset$ or $\{\}$ is the (unique) set that has no elements

Number of elements

$$\#\{4, 3, 3, 2, 1, 2\} = \#\{1, 2, 3, 4\} = 4$$

$$\#\emptyset = 0$$

Review exercises

Exercise 1

Write down all subsets of $\{1\}$ and of $\{1, 2\}$

Exercise 2

Are the following set inclusions true or false?

(a) $\{1, 3, 5, 7\} \subseteq \{7, 6, 5, 4, 3, 2, 6, 8\}$

(b) $\{1, 3, 5, 7\} \subseteq \{7, 4, 1, 6, 5, 4, 4, 3\}$

Review exercises

Exercise 1

Write down all subsets of $\{1\}$ and of $\{1,2\}$

of $\{1\}$: $\emptyset$ $\{1\}$

note that $\emptyset \subseteq \{1\}$

of $\{1,2\}$: $\emptyset$ $\{1\}$ $\{2\}$ $\{1,2\}$

note that $\emptyset \subseteq \{1,2\}$

Exercise 2

Are the following set inclusions true or false?

(a) $\{1,3,5,7\} \subseteq \{7,6,5,4,3,2,6,8\}$

false

(b) $\{1,3,5,7\} \subseteq \{7,4,1,6,5,4,4,3\}$

true

Sets: definition and notation

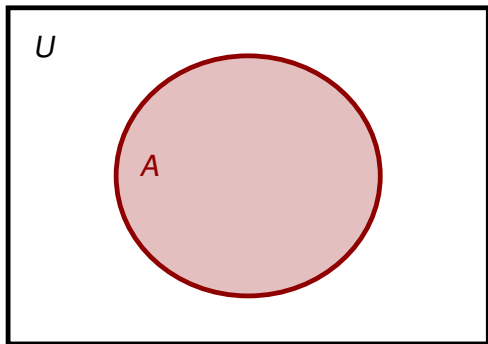
Fundamental set operations

Venn diagrams and partitions

The algebra of sets

Universe

All our sets are subsets of a **universe** U



Example

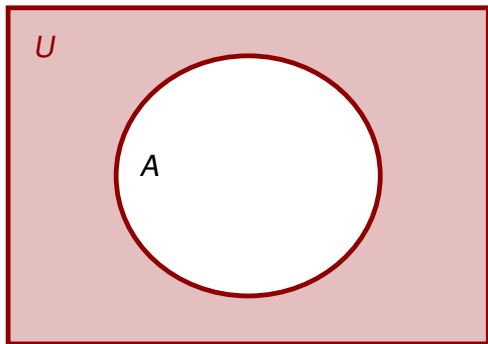
U : all 4-letter words

A : words with a

U defines the *context* we work in to avoid Russell's Paradox

Complement

Complement of A



Example

U : all 4-letter words

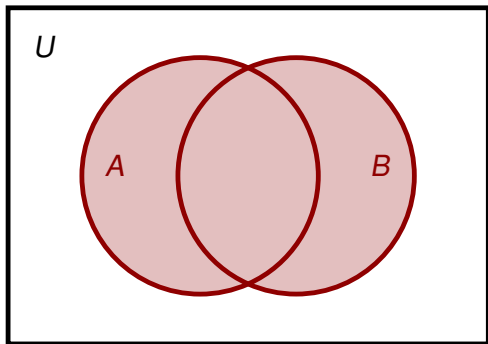
A : words with a

A' :
words with *no* a

$$A' := \{x \in U : x \notin A\}$$

Union and intersection

Union of A and B



Example

U : all 4-letter words

A : words with a

B : words with b

$A \cup B$:

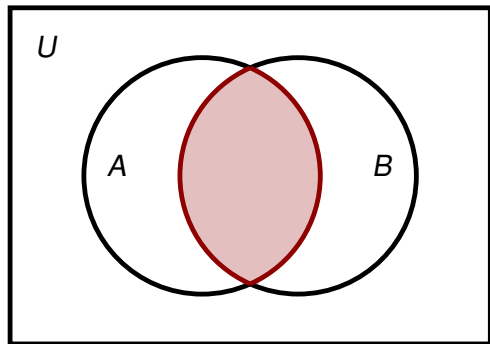
words with a or b

(inclusive or)

$$A \cup B := \{x \in U : x \in A \text{ or } x \in B\}$$

Union and intersection

Intersection of A and B



Example

U : all 4-letter words

A : words with a

B : words with b

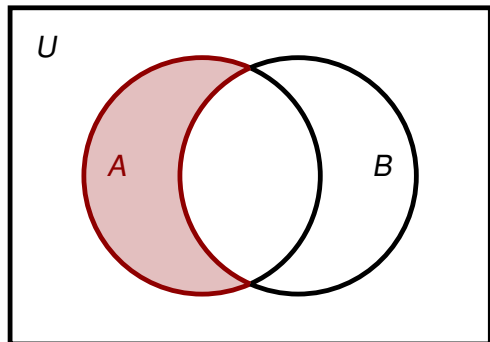
$A \cap B$:

words with a *and* b

$$A \cap B := \{x \in U : x \in A \text{ and } x \in B\}$$

Set-theoretic and symmetric difference

A minus B



Example

U : all 4-letter words

A : words with a

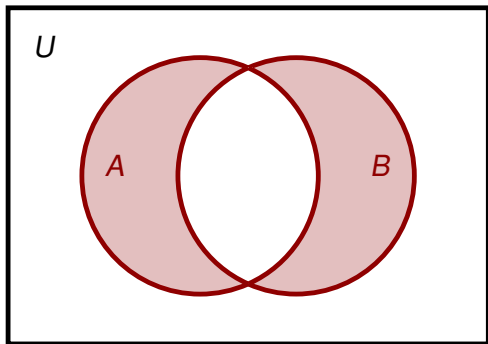
B : words with b

$A \setminus B$:
words with a *but no* b

$$A \setminus B := \{x \in A : x \notin B\} = A \cap B'$$

Set-theoretic and symmetric difference

Symmetric difference of A and B



Example

U : all 4-letter words

A : words with a

B : words with b

$A \triangle B$:

words with a *or* b

but not both

(exclusive or)

$$\begin{aligned} A \triangle B &:= (A \cup B) \cap (A \cap B)' \\ &= (A \setminus B) \cup (B \setminus A) \end{aligned}$$

Review exercises

Exercise

We are given the data:

$$A := \{0, 1, 2, 3\}$$

$$B := \{2, 3, 4, 5\}$$

The universe is $\mathbb{N}$

Determine the following sets:

$$A \cup B =$$

$$A \setminus B =$$

$$A \cap B =$$

$$B \setminus A =$$

$$A' =$$

$$A \triangle B =$$

$$B' =$$

Review exercises

Exercise

We are given the data:

$$A := \{0, 1, 2, 3\}$$

$$B := \{2, 3, 4, 5\}$$

The universe is $\mathbb{N}$

Determine the following sets:

$$A \cup B = \{0, 1, 2, 3, 4, 5\}$$

$$A \setminus B = \{0, 1\}$$

$$A \cap B = \{2, 3\}$$

$$B \setminus A = \{4, 5\}$$

$$A' = \{4, 5, 6, \dots\}$$

$$A \triangle B = \{0, 1, 4, 5\}$$

$$B' = \{0, 1\} \cup \{6, 7, 8, \dots\}$$

break

Sets: definition and notation

Fundamental set operations

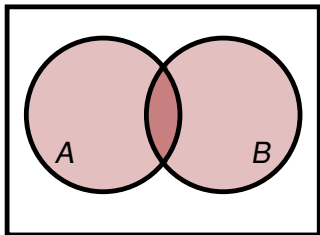
Venn diagrams and partitions

The algebra of sets

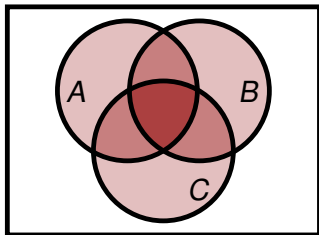
Venn diagrams

Venn diagram: abstract visualisation of relations between sets

2 sets $\Rightarrow$ 4 regions

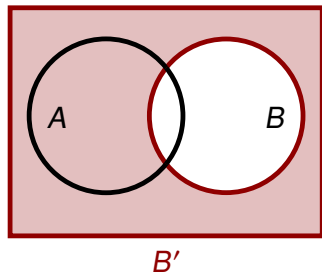
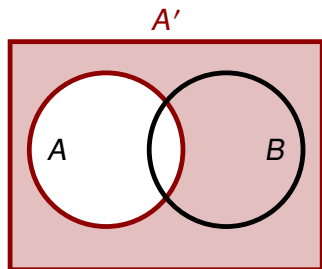


3 sets $\Rightarrow$ 8 regions

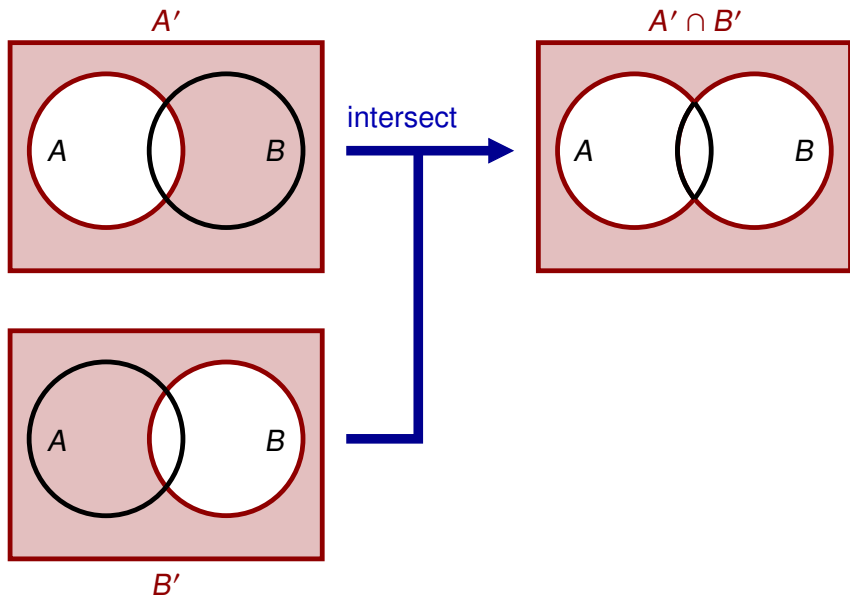


Every region in a Venn diagram can be empty or non-empty

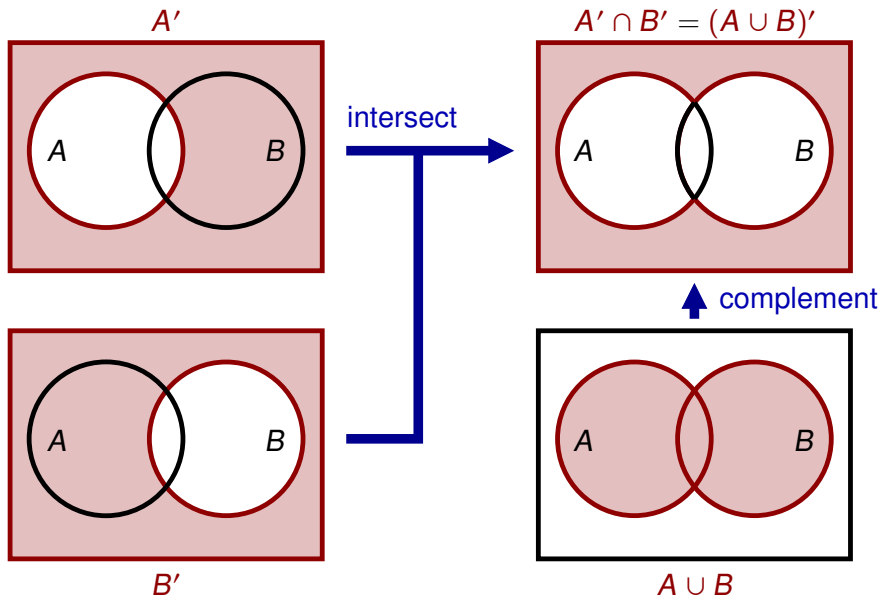
Venn diagrams: De Morgan's law $A' \cap B' = (A \cup B)'$



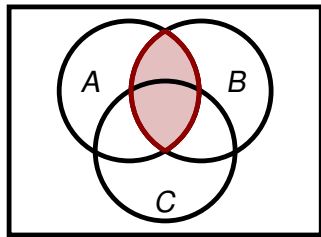
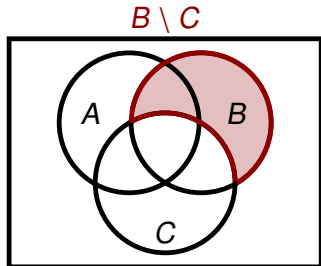
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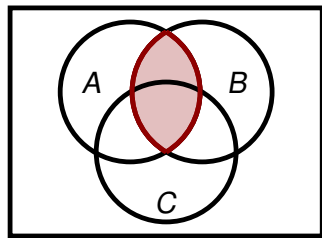
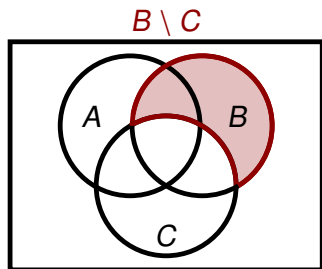


Venn diagrams: $A \cap (B \setminus C) = (A \cap B) \setminus C$



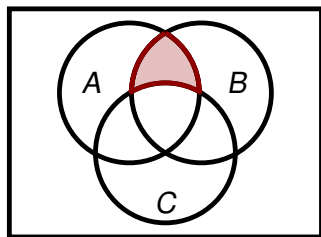
$A \cap B$

Venn diagrams: $A \cap (B \setminus C) = (A \cap B) \setminus C$



$A \cap B$

intersect with A
take away C



$$A \cap (B \setminus C) = (A \cap B) \setminus C$$

Venn diagrams: counting elements

Data

U has 15 elements

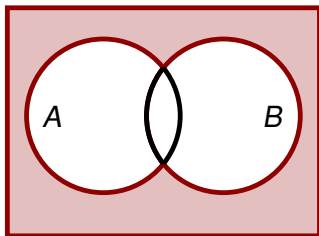
$A \cap B$ has 3 elements

A has 10 elements

B has 6 elements

Problem

Determine $\#(A \cup B)'$



Venn diagrams: counting elements

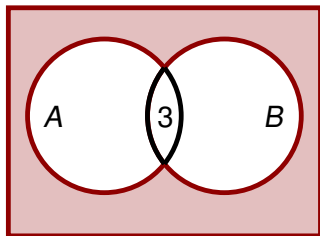
Data

U has 15 elements

$A \cap B$ has 3 elements ✓

A has 10 elements

B has 6 elements



Problem

Determine $\#(A \cup B)'$

Venn diagrams: counting elements

Data

U has 15 elements

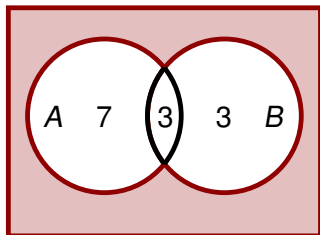
$A \cap B$ has 3 elements ✓

A has 10 elements ✓

B has 6 elements ✓

Problem

Determine $\#(A \cup B)'$



Venn diagrams: counting elements

Data

U has 15 elements ✓

$A \cap B$ has 3 elements ✓

A has 10 elements ✓

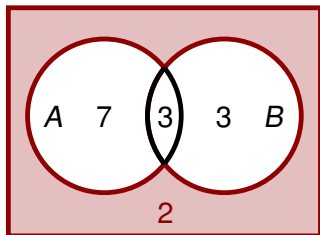
B has 6 elements ✓

Problem

Determine $\#(A \cup B)'$

Answer

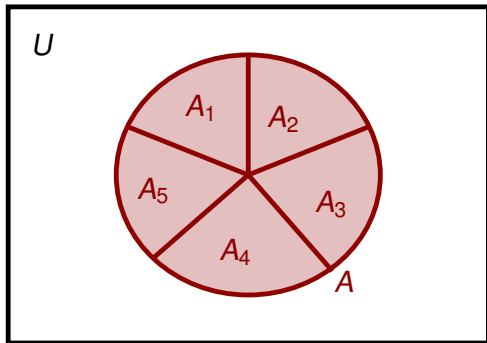
$$\#(A \cup B)' = 2$$



Partitions

A **partition** of A is a collection $\{A_1, \dots, A_n\}$ of **parts** $A_i \subseteq A$ such that

- ▶ the parts are *not empty*
- ▶ the parts *do not overlap* (are *disjoint*)
- ▶ the parts together *cover* A



Partitions

A **partition** of A is a collection $\{A_1, \dots, A_n\}$ of **parts** $A_i \subseteq A$ such that

- ▶ the parts are *not empty*
- ▶ the parts *do not overlap* (are *disjoint*)
- ▶ the parts together *cover* A

Rule of sum for partitions

$$\#A = \#A_1 + \#A_2 + \dots + \#A_n$$

“ A is the sum of the parts”

Example

$$\{0, 1, 2, \dots, 9\} \Rightarrow \text{partition } \{\{2, 4, 6, 8\}, \{0, 1, 9\}, \{3, 5, 7\}\}$$

$$10 = 4 + 3 + 3$$

Review exercises

Exercise 1

Verify using Venn diagrams that
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Exercise 2

We are given the data:

$$\#U = 28$$

$$\#A = \#B = \#C = 12$$

$$\#(A \cap B) = \#(A \cap C) = \#(B \cap C) = 4$$

$$\#(A \cap B \cap C) = 1$$

Use Venn diagrams to determine

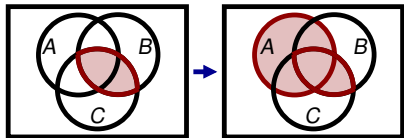
$$\#(A \cup B \cup C)$$

$$\#(B \cup C)'$$

Review exercises

Exercise 1

Verify using Venn diagrams that
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$



Exercise 2

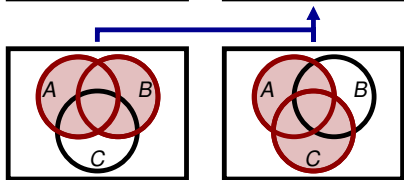
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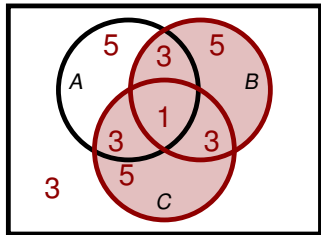
$$\#(A \cap B \cap C) = 1$$



Use Venn diagrams to determine

$$\#(A \cup B \cup C) = 25$$

$$\#(B \cup C)' = 8$$



Sets: definition and notation

Fundamental set operations

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The algebra of sets

The algebra of sets

Commutativity

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Complement

$$A \cap A' = \emptyset$$

$$A \cup A' = U$$

Idempotence

$$A \cap A = A$$

$$A \cup A = A$$

De Morgan

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

Associativity

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

Identity

$$A \cap U = A$$

$$A \cup \emptyset = A$$

Domination

$$A \cap \emptyset = \emptyset$$

$$A \cup U = U$$

Involution

$$(A')' = A$$

Definitions

$$A \setminus B = A \cap B'$$

$$A \triangle B = (A \cup B) \cap (A \cap B)'$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Repeated unions and intersections

Commutativity

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Associativity

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

Reorder rule

In a repeated union or repeated intersection you can omit brackets and change the order

Examples

- ▶ $A \cap (B \cap C) = A \cap B \cap C$
- ▶ $A \cap (B \cap C) = (B \cap A) \cap C$
- ▶ $(A \cup B) \cup (C \cup D) = D \cup C \cup B \cup A$
- ▶ $(A \cup B) \cup (C \cup D) = (C \cup A) \cup (D \cup B)$

Checking set equality using the algebra of sets

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Checking set equality using the algebra of sets

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Solution

$$\begin{aligned}(A' \cap B)' \cup (C \cap A')' &= ((A')' \cup B') \cup (C' \cup (A')') && \text{(De Morgan)} \\ &= (A \cup B') \cup (C' \cup A) && \text{(involution)} \\ &= (A \cup A) \cup (B' \cup C') && \text{(reorder)} \\ &= A \cup (B' \cup C') && \text{(idempotence)} \\ &= A \cup (B \cap C)' && \text{(De Morgan)}\end{aligned}$$

Review exercises

Exercise

Check that $(A' \cap (A \cup B'))' = A \cup B$ using the algebra of sets

Exercise

Check that $(A' \cap (A \cup B'))' = A \cup B$ using the algebra of sets

Solution

$$\begin{aligned}(A' \cap (A \cup B'))' &= (A')' \cup (A \cup B')' && \text{(De Morgan)} \\ &= (A')' \cup (A' \cap (B')') && \text{(De Morgan)} \\ &= A \cup (A' \cap B) && \text{(involution)} \\ &= (A \cup A') \cap (A \cup B) && \text{(distributivity)} \\ &= U \cap (A \cup B) && \text{(complement)} \\ &= A \cup B && \text{(identity)}\end{aligned}$$